

Exercise 1

Cross entropy loss with respect to logits.

$\hat{y}_i$: i -th softmax element

l : sum of exponentials of all logits

z_i : softmax input

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

$$\bullet \frac{\partial \hat{y}_i}{\partial l_n} \stackrel{i=n}{=} \frac{e^{z_i} \sum e^{z_j} - e^{z_i} e^{z_n}}{(\sum e^{z_j})^2} = \frac{e^{z_i}}{\sum e^{z_j}} \left(1 - \frac{e^{z_n}}{\sum e^{z_j}}\right) = \hat{y}_i (1 - \hat{y}_n) = \hat{y}_n (1 - \hat{y}_i)$$

$$\bullet \frac{\partial \hat{y}_i}{\partial l_n} \stackrel{i \neq n}{=} \frac{0 \cdot \sum e^{z_j} - e^{z_i} e^{z_n}}{(\sum e^{z_j})^2} = -\frac{e^{z_i}}{\sum e^{z_j}} \cdot \frac{e^{z_n}}{\sum e^{z_j}} = -\hat{y}_i \hat{y}_n$$

$$\text{Thus, we get } \frac{\partial \hat{y}_i}{\partial l_n} = \begin{cases} \hat{y}_n (1 - \hat{y}_n), & i = n \\ -\hat{y}_i \hat{y}_n, & i \neq n \end{cases}$$

$\Rightarrow$ Cross entropy loss - Derivative

$$\bullet l_i = -y_i \log(\hat{y}_i)$$

$$\bullet L = -\sum_i y_i \log(\hat{y}_i)$$

$$\begin{aligned} \frac{\partial L}{\partial l_n} &= -\sum_i y_i \cdot \frac{\partial \log(\hat{y}_i)}{\partial l_n} = -\sum_i y_i \cdot \frac{\partial \log(\hat{y}_i)}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial l_n} = \\ &= -\sum_i y_i \cdot \frac{1}{\hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial l_n} = -\sum_i \frac{y_i}{\hat{y}_i} \cdot \boxed{\frac{\partial \hat{y}_i}{\partial l_n}} \rightarrow \text{softmax derivative} \end{aligned}$$

Now we have:

$$\begin{aligned} \bullet i=n \quad \frac{\partial L}{\partial l_n} &= -\sum_i \frac{y_i}{\hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial l_n} = -\sum_i \frac{y_i}{\hat{y}_i} \cdot [\hat{y}_n (1 - \hat{y}_n)] = \\ &= -\sum_{i=n} \frac{y_i}{\hat{y}_i} \cdot \hat{y}_n (1 - \hat{y}_n) = -y_n + y_n \hat{y}_n \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \bullet i \neq n \quad \frac{\partial L}{\partial l_n} &= -\sum_i \frac{y_i}{\hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial l_n} = -\sum_i \frac{y_i}{\hat{y}_i} [-\hat{y}_i \hat{y}_n] = \sum_{i \neq n} \frac{y_i}{\hat{y}_i} \cdot \hat{y}_i \hat{y}_n = \\ &= \sum_{i \neq n} y_i \hat{y}_n \quad \textcircled{2} \end{aligned}$$

From $\textcircled{1}$ & $\textcircled{2}$ we end up with: $\frac{\partial L}{\partial l_n} = -y_n + \sum_i y_i \hat{y}_n = -y_n + 1 \cdot \hat{y}_n \Rightarrow$

All y components are 0, except for one of them that is 1. So, the sum in all previous steps becomes 1 by $\hat{y}_n$.

$$\boxed{\frac{\partial L}{\partial l_n} = \hat{y}_n - y_n}$$